

Network Performance Models

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CS4226: Internet Architecture

Network Performance Metrics

□ Link Rate/Bandwidth/Capacity

- ❖ Cable - 1~100 Mbps, Fiber optics - 1~10 Gbps
- ❖ Ethernet - 3Mbps to 100 Gbps
- ❖ How many bits can be “pushed” onto a link per unit time

□ Throughput

- ❖ Throughput of a TCP/UDP flow
- ❖ How many bits can be communicated per unit time

Network Performance Metrics

□ End-to-end delay

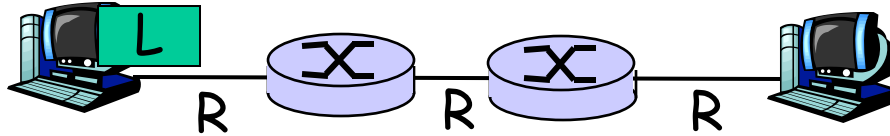
❖ Components:

- processing + queueing delay + transmission + propagation

❖ An example:

- if a source sends a packet of L bits over a link with rate R bits/sec, the time to transmit the packet is ...

Packet-switching: store-and-forward



- ❖ takes L/R seconds to transmit (push out) packet of L bits on to link at R bps
- ❖ **store and forward:** entire packet must arrive at router before it can be transmitted on next link
- ❖ delay = $3L/R$ (assuming zero propagation delay)

Example:

- $L = 7.5$ Mbits
- $R = 1.5$ Mbps
- transmission delay = 15 seconds

} more on delay shortly ...

Network Performance Metrics

□ End-to-end delay

❖ Components:

- processing + queueing delay + transmission + propagation

❖ Relationship to

- bandwidth, packet size, distance, propagation speed, and queue size

□ Response time

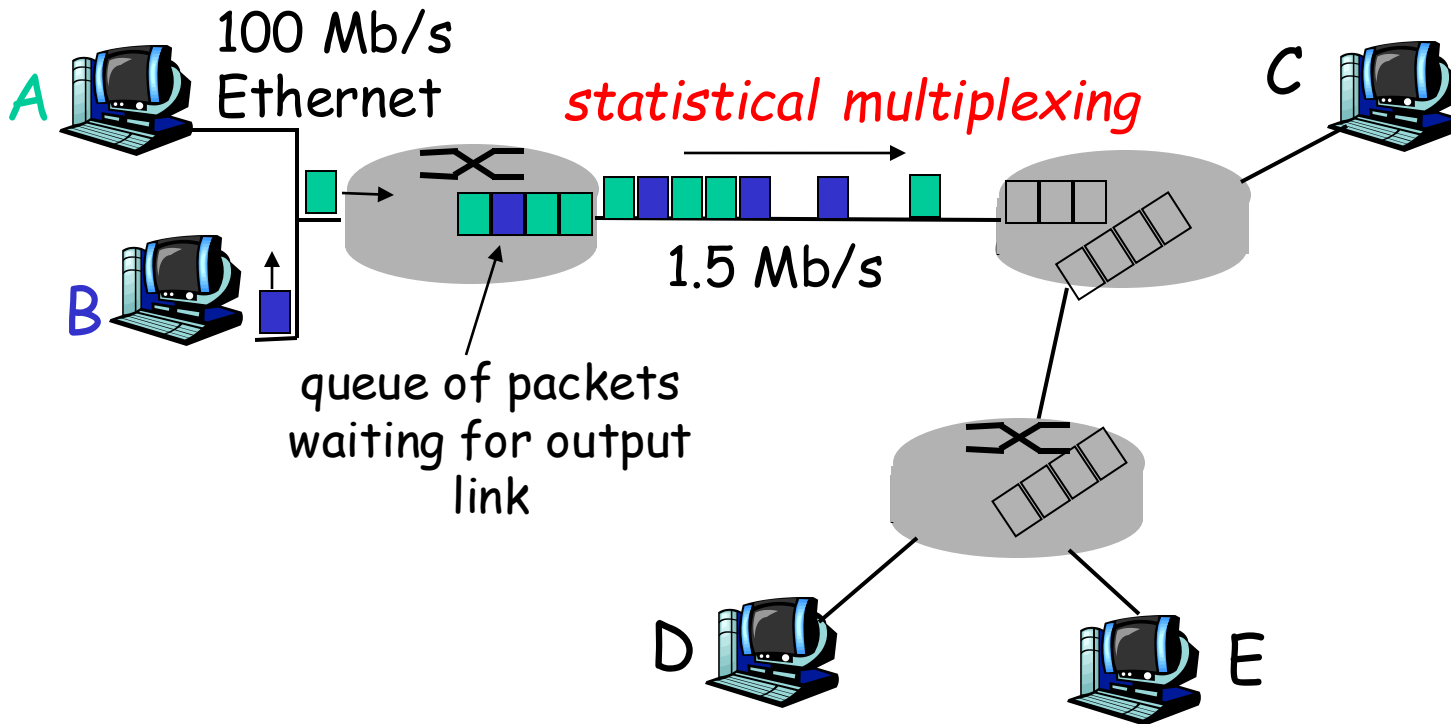
□ Other metrics

- ❖ Dropping probability
- ❖ Utilization

Questions

- ❑ What is the origin of queueing delay?
- ❑ How do we calculate the queueing delay?
 - ❖ what does it depend on?
- ❑ Is there a relationship between queueing delay and throughput?
 - ❖ positively or negatively correlated?

Packet Switching: Statistical Multiplexing



- ❖ sequence of A & B packets has no fixed timing pattern
 - bandwidth shared on demand: *statistical multiplexing*.
- ❖ TDM: each host gets same slot in revolving TDM frame.

Packet switching versus circuit switching

Packet switching allows more users to use network!

Example:

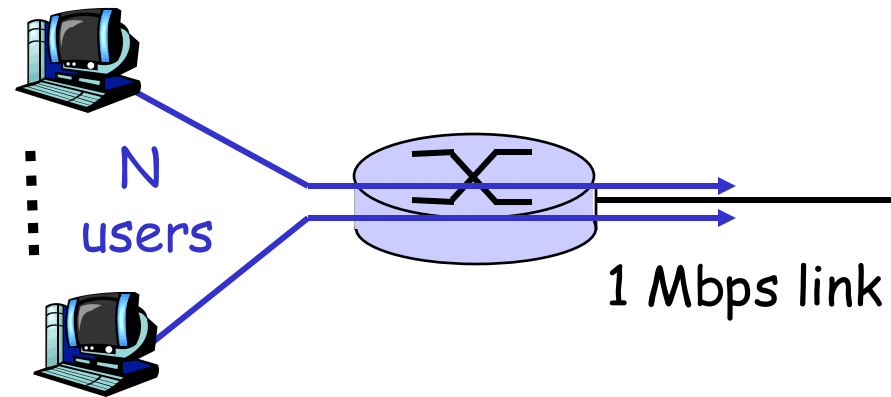
- 1 Mb/s link
- each user:
 - 100 kb/s when "active"
 - active 10% of time

❖ *circuit-switching:*

- 10 users

❖ *packet switching:*

- with 35 users, probability
> 10 active at same time
is less than .0004



Q: how did we get value 0.0004?

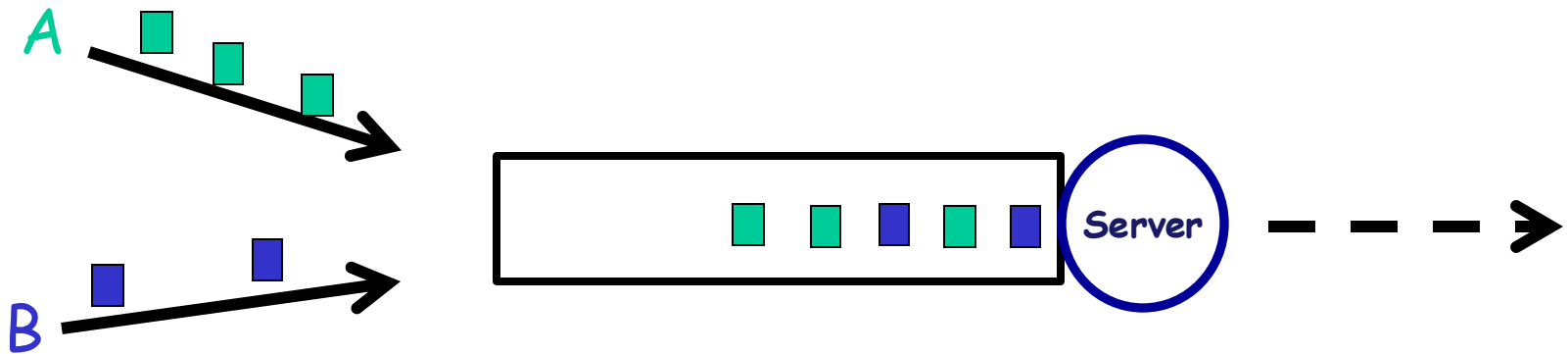
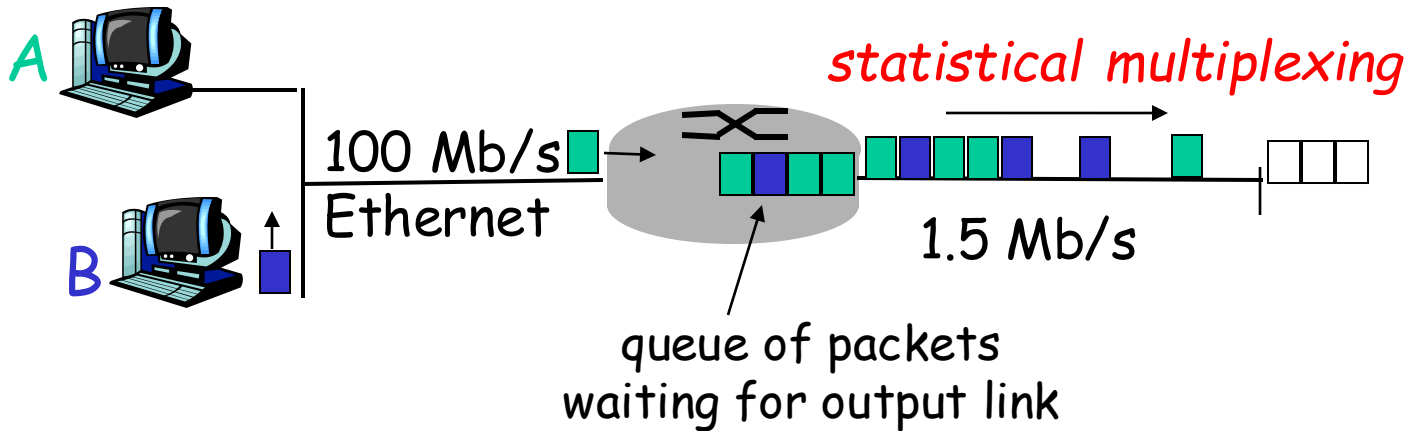
Q: what happens if > 35 users ?

Packet switching versus circuit switching

Is packet switching a "slam dunk winner?"

- ❖ great for (asynchronous) bursty data
 - resource sharing
 - simpler, no call setup
- ❖ **excessive congestion:** packet delay and loss
 - protocols needed for reliable data transfer, congestion control
- ❖ human analogies of
 - reserved resources (circuit switching)
 - versus on-demand allocation (packet-switching)

A single "server" queueing model

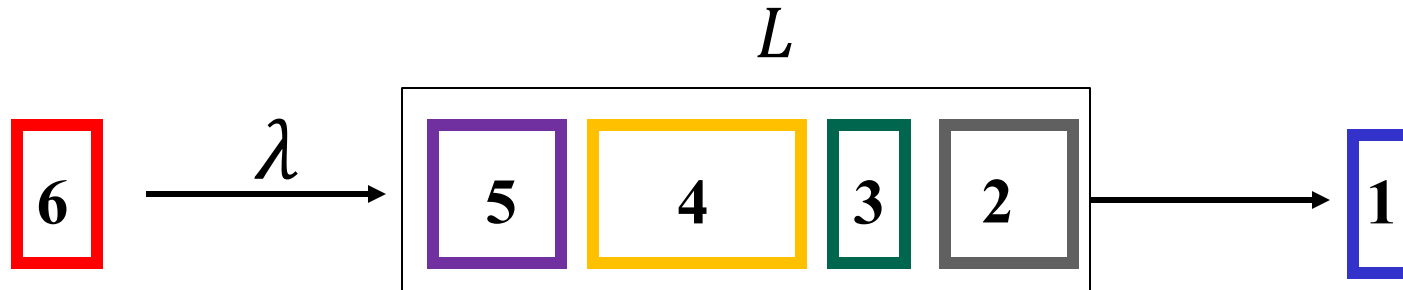


A "server" with an infinity queue
and a capacity of 1.5 Mb/s

Little's Law

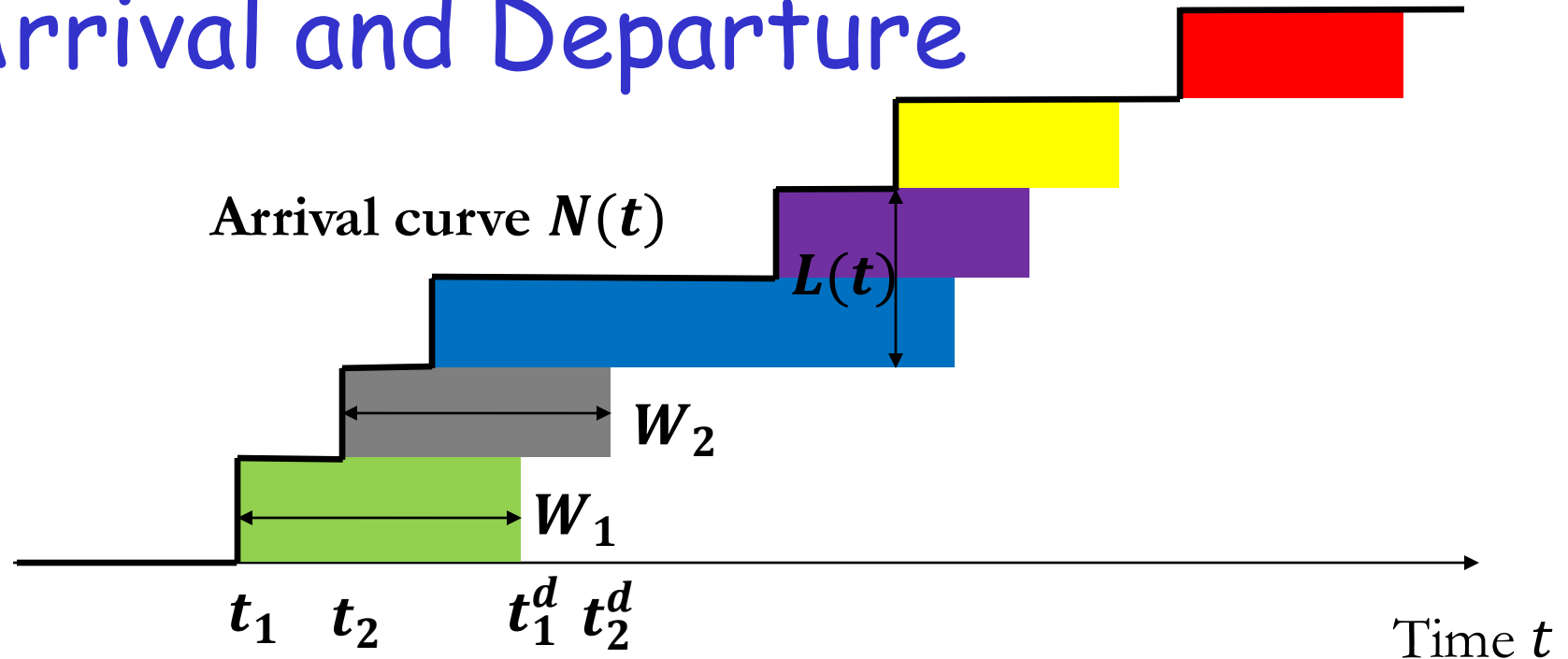
- ❑ SoC admits 1500 UG students/year on average and has on average 6000 UG students at any time
- ❖ How long does a student expect to spend before getting a bachelor's degree?
- ❑ How about this?
 - ❖ If avg. queue length is 5 and packets arrive at a rate of 100/sec, what's the avg. queueing delay?
- Avg. number of customers in system
= arrival rate $\times$ avg. sojourn time

An Abstract System



- Arrival rate: λ
- # of customers/packets in the system: L
- Sojourn time W_i for the i th customer
 - ❖ the time a customer spent in the system

Arrival and Departure



- t_i and t_i^d : arrival and departure time of i th customer
- $N(t)$: # of customers arrived till time t
- $L(t)$: # of customers in the system at time t
- W_i : sojourn time of customer
 - ❖ Fact: $t_i^d = t_i + W_i$

Little's Law : $L = \lambda W$

- Arrival rate into the system λ :

$$\lambda \stackrel{\text{def}}{=} \lim_{t \rightarrow \infty} \frac{N(t)}{t}$$

- Average sojourn time W :

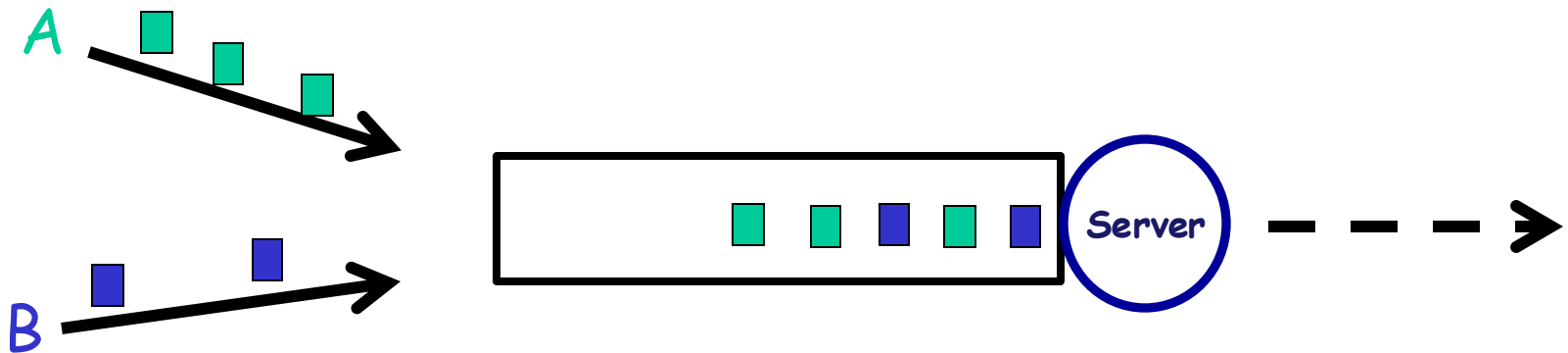
$$W \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n W_j$$

- Time average # of customers in system L :

$$L \stackrel{\text{def}}{=} \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t L(s) ds$$

A single "server" queueing model

- If the avg. queue length is 5 packets, and on average 100 packets arrive per second, what is the avg. queueing delay for packet?
- What can be said beyond the mean?



A "server" with an infinity queue
and a capacity of 1.5 Mb/s